ALEXANDER SAINSBURY

DATA ANALYTICS ENGINEER
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SUMMARY

Data-focused professional specialising in transforming raw, multi-source datasets into structured, analysis-ready data pipelines and business insights.
Experienced in Python, SQL, and Excel-based ETL workflows, with a strong focus on data cleaning, modelling, and visualisation. Demonstrated ability to automate manual processes, improve data integrity, and deliver actionable insights through dashboards and analytics.

KEY PROJECTS

Finance Data Pipeline & Dashboarding

Tools: Excel (Power Query), Tableau

Overview
Designed and implemented an end-to-end data pipeline to consolidate and analyse multi-source financial datasets (Visa and Amex), each with inconsistent formats and Japanese-language fields.

Approach
- Built an automated ETL workflow using Power Query to ingest and transform monthly CSV files
- Developed a translation mapping system using XLOOKUP, UNIQUE, and SORT to standardise Japanese data into English
- Cleaned and structured data by:
  - removing duplicates
  - resolving null values
  - enforcing a consistent schema
- Created a unified dataset suitable for downstream analysis and reporting

Output
- Interactive Tableau dashboards showing:
  - spending distribution across accounts
  - monthly and yearly trends
  - geographic breakdowns

Impact
- Reduced manual data preparation time from hours to minutes
- Enabled real-time analysis of newly updated financial data
- Identified actionable spending patterns, reducing monthly expenditure from ¥80,000 to under ¥50,000 (~37.5% reduction)

Web-Scraped Dataset ETL Pipeline

Tools: Python (pandas, NumPy), SQLite

Overview

Developed a modular ETL pipeline to extract, transform, and store semi-structured web data into a structured analytical dataset.

Approach

- Extracted data from nested HTML tables using web scraping techniques
- Implemented a layered ETL architecture (extract / transform / load)
- Engineered structured features from raw attributes
- Normalised data schema for relational storage
- Stored outputs in CSV and SQLite databases for reproducibility

Output

- Clean, quarriable dataset for analysis
- Reusable pipeline for processing similar web-based data sources

Impact

- Demonstrated scalable data pipeline design principles
- Improved data accessibility and reproducibility for analysis workflows

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Scientific Simulation: Diffusion-Limited Aggregation

Tools: Python (NumPy, Matplotlib)

Overview

Built a 3D simulation model to analyse stochastic particle aggregation and fractal growth patterns.

Approach

- Implemented a Monte Carlo random walk algorithm in three dimensions
- Simulated particle interactions and cluster formation
- Dynamically adjusted simulation boundaries as structures grew

Output

- 3D visualisations of particle aggregation structures
- Exportable datasets for further analysis

Relevance

- Demonstrates algorithmic thinking, numerical modelling, and performance optimisation
- Applicable to data engineering through handling of large datasets and computational workflows

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Media Performance Modelling (Ongoing)

Tools: Python, statistical modelling

Overview

Analysing social media performance data to identify trends and forecast growth.

Approach

- Applied statistical and probability models to forecast engagement trends
- Identified key features associated with high-performing content
- Translated stakeholder requirements into measurable KPIs

Impact

- Supporting data-driven decision making for content strategy
- Developing real-world experience in stakeholder collaboration and requirements gathering

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Technical Skills

Programming & Data

- Python (pandas, NumPy, matplotlib, SciPy, BeautifulSoup)
- SQL (joins, aggregations, window functions, CTEs)

Data Engineering

- ETL pipeline development
- Data cleaning, transformation, and validation
- Data modelling and schema design
- Relational databases (SQLite)

Visualisation & BI

- Tableau (interactive dashboards)
- Power BI (working knowledge)

Tools

- Excel (Power Query, advanced formulas)
- GitHub, VS Code, Jupyter Notebook

Key Strengths

- Designing end-to-end data pipelines from raw data to insight
- Transforming inconsistent datasets into reliable, structured formats
- Communicating insights clearly through visual dashboards
- Applying analytical thinking to solve real-world problems

Closing Statement

This portfolio demonstrates practical experience in building scalable data workflows, transforming complex datasets, and delivering business-focused insights.